Problem Set 1: Question 1 and 2:

Problem Set 1b: Question 2.1 and 2.3

| | | | | | | |
|---|---|---|---|---|---|---|
| 1*a*. | $0 \cdot 0 = 0$ | 5*a*. | $x \cdot 0 = 0$ | 10*a*. | $x \cdot y = y \cdot x$ | *Commutative* |
| 1*b*. | $1 + 1 = 1$ | 5*b*. | $x + 1 = 1$ | 10*b*. | $x + y = y + x$ | |
| 2*a*. | $1 \cdot 1 = 1$ | 6*a*. | $x \cdot 1 = x$ | 11*a*. | $x \cdot (y \cdot z) = (x \cdot y) \cdot z$ | *Associative* |
| 2*b*. | $0 + 0 = 0$ | 6*b*. | $x + 0 = x$ | 11*b*. | $x + (y + z) = (x + y) + z$ | |
| 3*a*. | $0 \cdot 1 = 1 \cdot 0 = 0$ | 7*a*. | $x \cdot x = x$ | 12*a*. | $x \cdot (y + z) = x \cdot y + x \cdot z$ | *Distributive* |
| 3*b*. | $1 + 0 = 0 + 1 = 1$ | 7*b*. | $x + x = x$ | 12*b*. | $x + y \cdot z = (x + y) \cdot (x + z)$ | |
| 4*a*. | If $x = 0$, then $\overline{x} = 1$ | 8*a*. | $x \cdot \overline{x} = 0$ | 13*a*. | $x + x \cdot y = x$ | *Absorption* |
| 4*b*. | If $x = 1$, then $\overline{x} = 0$ | 8*b*. | $x + \overline{x} = 1$ | | | |
| | | 9. | $\overline{\overline{x}} = x$ | | | |

| | | |
|---|---|---|
| 13*b*. | $x \cdot (x + y) = x$ | |
| 14*a*. | $x \cdot y + x \cdot \overline{y} = x$ | *Combining* |
| 14*b*. | $(x + y) \cdot (x + \overline{y}) = x$ | |
| 15*a*. | $\overline{x \cdot y} = \overline{x} + \overline{y}$ | *DeMorgan's theorem* |
| 15*b*. | $\overline{x + y} = \overline{x} \cdot \overline{y}$ | |
| 16*a*. | $x + \overline{x} \cdot y = x + y$ | |
| 16*b*. | $x \cdot (\overline{x} + y) = x \cdot y$ | |
| 17*a*. | $x \cdot y + y \cdot z + \overline{x} \cdot z = x \cdot y + \overline{x} \cdot z$ | *Consensus* |
| 17*b*. | $(x + y) \cdot (y + z) \cdot (\overline{x} + z) = (x + y) \cdot (\overline{x} + z)$ | |

Problem Set 2: Question 6

Figure: 3.66

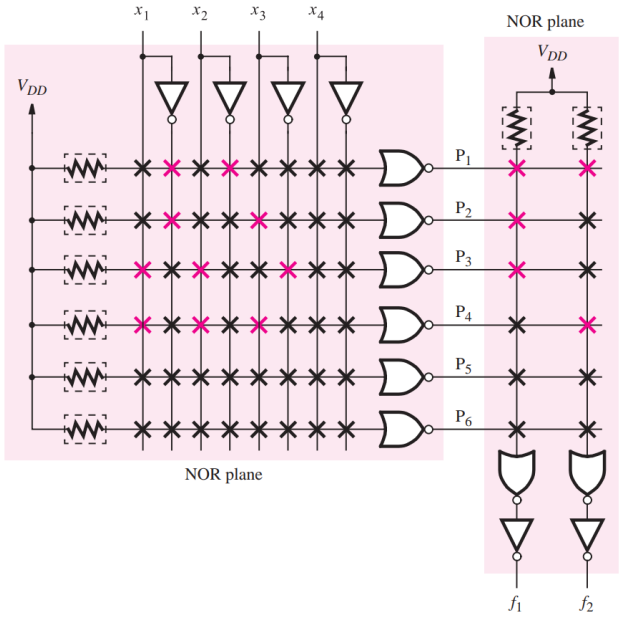


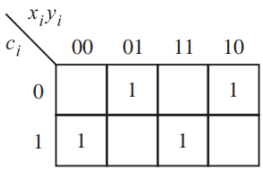

**Figure 3.66** A NOR-NOR PLA used for sum-of-products.

Problem Set 4: Question 5

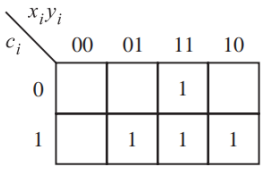
Figure: 5.4

| $c_i$ | $x_i$ | $y_i$ | $c_{i+1}$ | $s_i$ |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 |

(a) Truth table

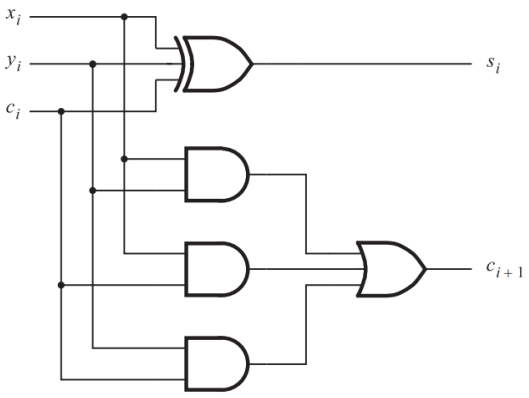



$s_i = x_i \oplus y_i \oplus c_i$




$c_{i+1} = x_iy_i + x_ic_i + y_ic_i$

(b) Karnaugh maps

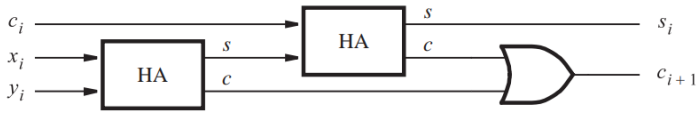



(c) Circuit

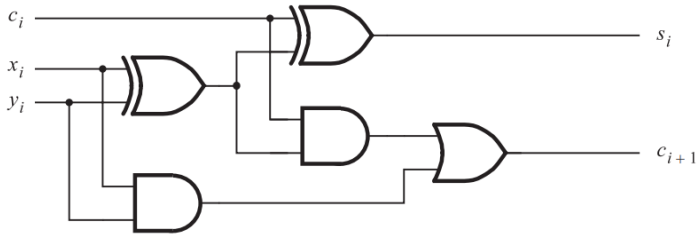
**Figure 5.4** Full-adder.

Problem Set 4: Question 5

Figure: 5.5




(a) Block diagram




(b) Detailed diagram

**Figure 5.5** A decomposed implementation of the full-adder circuit.

Problem Set 5: Question 10

Figure: 6.55

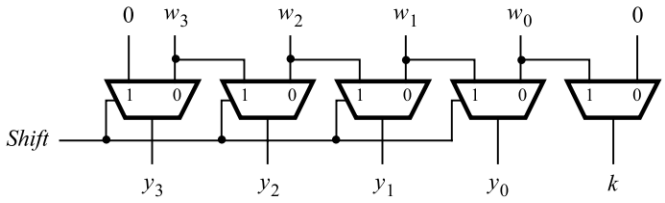
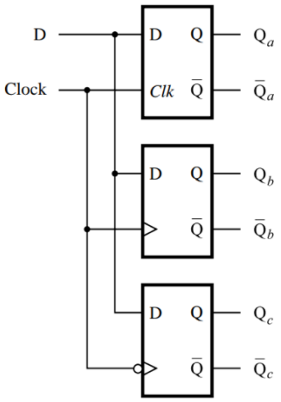


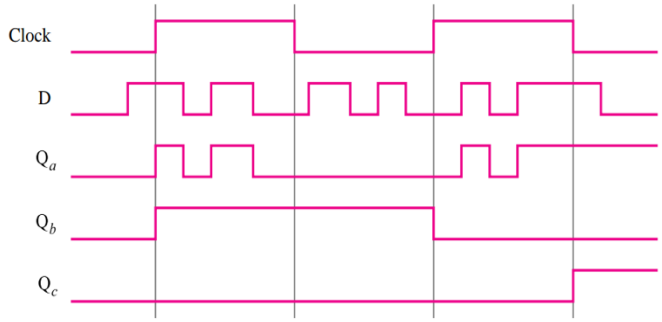

**Figure 6.55** A shifter circuit.

Problem Set 6: Question 1

Figure: 7.12




(a) Circuit




(b) Timing diagram

**Figure 7.12** Comparison of level-sensitive and edge-triggered D storage elements.

Problem Set 7: Question 12

Figure 8.58

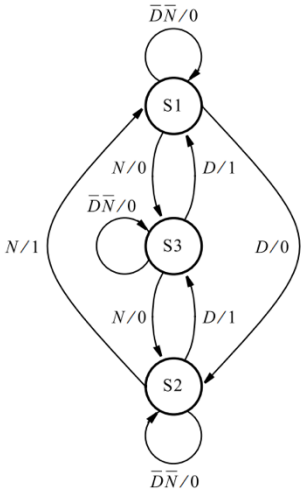



**Figure 8.58** Mealy-type FSM for Example 8.6.

Figure 8.3

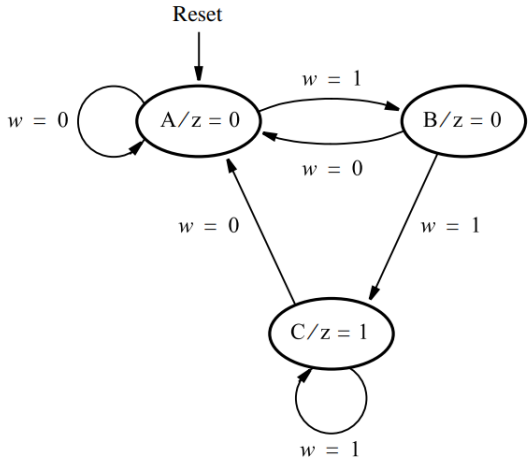



**Figure 8.3** State diagram of a simple sequential circuit.

Problem Set 7: Question 12

Figure 8.33

```
ARCHITECTURE Behavior OF simple IS
    TYPE State_type IS (A, B, C) ;
    SIGNAL y_present, y_next : State_type ;
BEGIN
    PROCESS ( w, y_present )
    BEGIN
        CASE y_present IS
            WHEN A =>
                IF w = '0' THEN
                    y_next <= A ;
                ELSE
                    y_next <= B ;
                END IF ;
            WHEN B =>
                IF w = '0' THEN
                    y_next <= A ;
                ELSE
                    y_next <= C ;
                END IF ;
            WHEN C =>
                IF w = '0' THEN
                    y_next <= A ;
                ELSE
                    y_next <= C ;
                END IF ;
        END CASE ;
    END PROCESS ;

    PROCESS (Clock, Resetn)
    BEGIN
        IF Resetn = '0' THEN
            y_present <= A ;
        ELSIF (Clock'EVENT AND Clock = '1') THEN
            y_present <= y_next ;
        END IF ;
    END PROCESS ;

    z <= '1' WHEN y_present = C ELSE '0' ;
END Behavior ;
```

**Figure 8.33** Alternative style of VHDL code for the FSM in Figure 8.3.